Summary in Graph

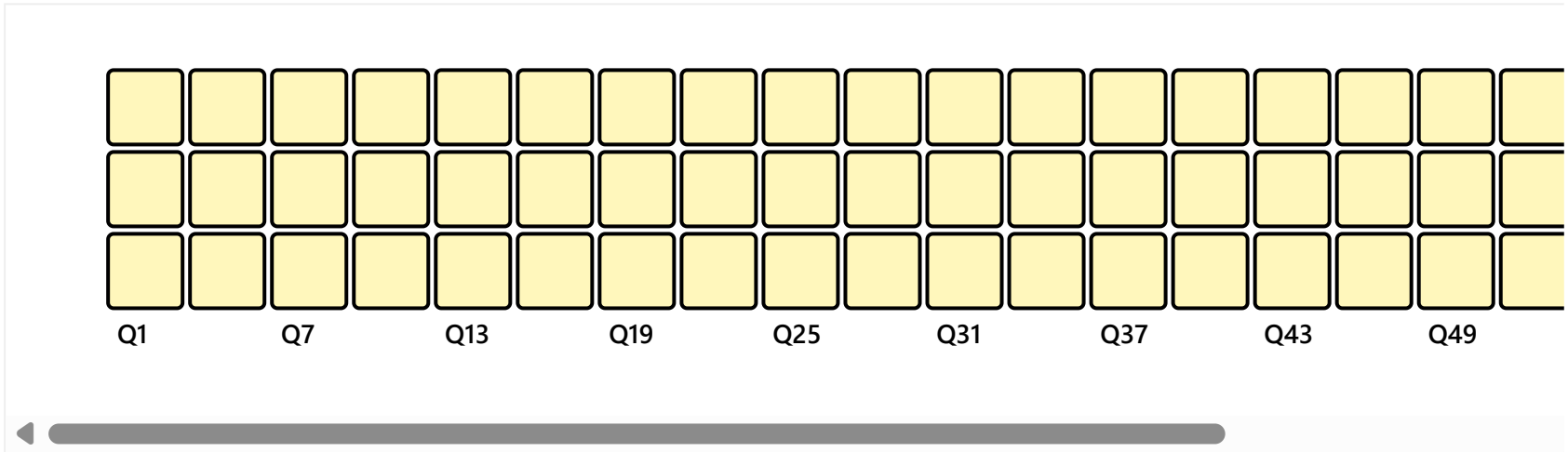
Exam Summary (GO Classes CS Test Series 2026 | MOCK GATE | Test 13)

Qs. Attempted:	0 0 + 0	Correct Marks:	0 0 + 0
Correct Attempts:	0 0 + 0	Penalty Marks:	0 0 + 0
Incorrect Attempts:	0 0 + 0	Resultant Marks:	0 0 + 0

Total Questions:	65 30 + 35
Total Marks:	100 30 + 70
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (73) out of 727 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Until Iran came along, India had never been _____ in Kabaddi.

A. defeated
B. defeating
C. defeat
D. defeatist

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 02sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

Which of the following forms is valid?

A. No P are M. No M are S. So, no S are P.
B. All M are P. All M are S. So, all S are P.
C. All P are M. All S are M. So, all S are P.
D. No M are P. Some S are M. So, some S are not P.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

There are eight bags of rice looking alike, seven of which have equal weight and one is slightly heavier. The weighing balance is of unlimited capacity.

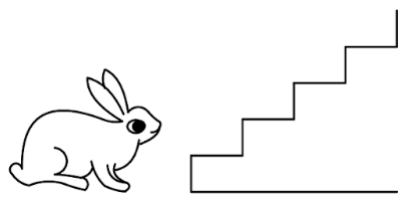
Using this balance, the minimum number of weighings required to identify the heavier bag is

A. 2
B. 3
C. 4
D. 8

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Buzz Bunny is hopping up and down a set of stairs, one step at a time. In how many ways can Buzz Bunny start on the ground, make a sequence of 6 hops, and end up back on the ground? (For example, one sequence of hops is up-up-down-down-up-down.)



A. 4
B. 5
C. 6
D. 8

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

A drawer contains red, green, blue, and white socks with at least 2 of each color. What is the minimum number of socks that must be pulled from the drawer to guarantee a matching pair?

A. 3
B. 4
C. 5
D. 8

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

The price of an item is 10% cheaper in an online store S compared to the price at another online store M. Store S charges ₹ 150 for delivery. There are no delivery charges for orders from the store M. A person bought the items from the store S and saved ₹ 100.

What is the price of the item at the online store S (in ₹) if there are no other charges than what is described above?

A. 2500
B. 2250
C. 1750
D. 1500

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

Five numbers $10, 7, 5, 4$ and 2 are to be arranged in a sequence from left to right following the directions given below:

1. No two odd or even numbers are next to each other.
2. The second number from the left is exactly half of the left-most number.
3. The middle number is exactly twice the right-most number.

Which is the second number from the right?

A. 2
B. 4
C. 7
D. 10

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type Award: 2 Penalty: 0.67 Analytical Aptitude

On Monday, Taye has $\$2$. Every day, he either gains $\$3$ or doubles the amount of money he had on the previous day. How many different dollar amounts could Taye have on Thursday, 3 days later?

A. 3
B. 4
C. 5
D. 6

Your Answer: Correct Answer: D Not Attempted Time taken: 00min 00sec Discuss

Q #9 Multiple Choice Type Award: 2 Penalty: 0.67 Quantitative Aptitude

On a 50-question multiple choice math contest, students receive 4 points for a correct answer, 0 points for an answer left blank, and -1 point for an incorrect answer. Jesse's total score on the contest was 99. What is the maximum number of questions that Jesse could have answered correctly?

A. 25
B. 27
C. 29
D. 31

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #10 Multiple Choice Type Award: 2 Penalty: 0.67 Verbal Aptitude

The importance of sleep is often overlooked by students when they are preparing for exams. Research has consistently shown that sleep deprivation greatly reduces the ability to recall the material learnt. Hence, cutting down on sleep to study longer hours can be counterproductive.

Which one of the following statements is the CORRECT inference from the above passage?

A. Sleeping well alone is enough to prepare for an exam. Studying has lesser benefit.
B. Students are efficient and are not wrong in thinking that sleep is a waste of time.
C. If a student is extremely well prepared for an exam, he needs little or no sleep.
D. To do well in an exam, adequate sleep must be part of the preparation.

Your Answer: Correct Answer: D Not Attempted Time taken: 00min 00sec Discuss

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Programming in C

Suppose a valid C program contains the statement

```
TREE = TRUNK;
```

Which of the following is true? (Don't assume anything about the style used in naming identifiers.)

A. TREE and TRUNK must both be #define.
B. TREE must be a #define but TRUNK doesn't have to be.

C. TREE and TRUNK both can be #define.
D. TREE cannot be a #define but TRUNK might be.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Programming in C

```
void swapAndPrint() {
    int list1[] = {5,3,7};
    int list2[] = {4, 8, 3};
    for (int i = 0; i < 3; ++i) {
        if (list1[i] > list2[i]) {
            int val = list1[i];
            list1[i] = list2[i];
            list2[i] = val;
        }
    }
    printf("%d", list2[1] + list2[2]);
}
```

What will be the output of swapAndPrint() function call?

Your Answer: | Correct Answer: 15 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

On a 64-bit system, which of the following C expressions is equivalent to the C expression $(x[2]+4)[3]$? Assume x is declared as int $**x$.

A. $*((*(x+16))+28)$
B. $*((*(x+2))+7)$
C. $**(x+28)$
D. $*(((*x)+2)+7)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the receiver receives the bits 0111110x0, what should it do? Assume there has not been any noise in the channel.

A. Remove the zero before the x, regardless of x.
B. Remove the zero before the x only if $x=0$.
C. Remove the zero before the x only if $x=1$.
D. Ignore the frame, since it must contain a bit error

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

P, Q, and R are the decimal integers corresponding to the 4-bit binary number 1100 considered in signed magnitude, 1's complement, and 2's complement representations, respectively. The 6-bit 2's complement representation of $(\mathrm{P}+\mathrm{Q}+\mathrm{R})$ is _____.

A. 110101
B. 110010
C. 111101
D. 111001

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

We have a relation $\mathrm{R}(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D})$. In each of the following questions, you are given a set F of functional dependencies, and a decomposition of R into two tables $R1$ and $R2$. In each case, determine whether the decomposition is lossy or lossless.

- $\mathrm{S1} : \mathrm{F} = \{\mathrm{BC} \rightarrow \mathrm{D}, \mathrm{C} \rightarrow \mathrm{D}\}$; decompose into $R1(A, B, C)$ and $R2(B, D)$.
- $\mathrm{S2} : \mathrm{F} = \{B \rightarrow C, C \rightarrow A, AC \rightarrow D\}$; decompose into $R1(A, B, C)$ and $R2(B, D)$.

Which of the following is correct?

A. $S1$ is a lossless decomposition and $S2$ is a lossless decomposition.
B. $S1$ is a lossless decomposition and $S2$ is a lossy decomposition.
C. $S1$ is a lossy decomposition and $S2$ is a lossless decomposition.
D. $S1$ is a lossy decomposition and $S2$ is a lossy decomposition.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

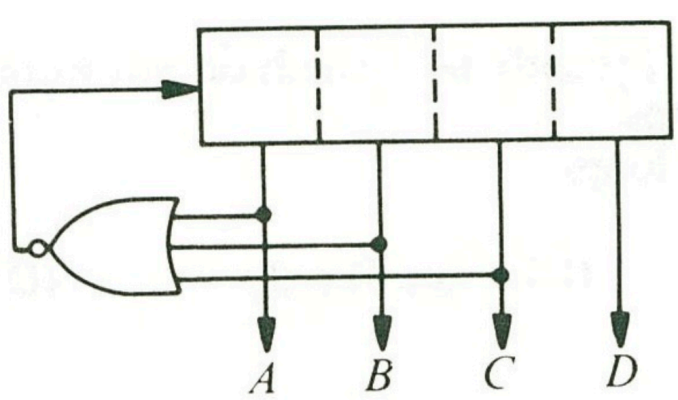
Assume that a data file has an index consisting of N items, where N is large. If a binary search of the index is used to find an item, then, of the following, which best approximates the mean number of comparisons required to locate a specific index entry?

A. $(\mathrm{N}+1)/2$
B. $\mathrm{N}(\mathrm{N}-1)/2$
C. $(\log_2 \mathrm{N}) - 1$
D. $\mathrm{N} \log_2 \mathrm{N}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The figure below shows a 4-bit, right shift register, and a NOR gate. If the register outputs $\{\mathrm{ABCD}\}$ at time 0 are 0110, then their values 4 clock pulses later are :



A. 0100
B. 0110
C. 0111
D. 1000

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider a relation $\mathrm{R}(\mathrm{A}, \mathrm{B})$ that contains 100 tuples, and a relation $\mathrm{S}(\mathrm{B}, \mathrm{C})$ that contains 80 tuples; Make no assumptions about keys $\&$ assume that the domain of all attributes is integers. Consider the following relational algebra expression:

$$\pi_{\mathrm{B}}\mathrm{R} - (\pi_{\mathrm{B}}\mathrm{R} - \pi_{\mathrm{B}}\mathrm{S})$$

Let m be the minimum and M be the maximum number of tuples that could be in the result of the above expression, then $m + \mathrm{M}$?

Your Answer: | Correct Answer: 80 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following function which takes node of a binary tree as an argument and tries to find something.

```
int findsomething(Node p) {
    if (p == NULL) return -1;
    if (p→left == NULL)
        return 1 + findsomething(p→right);
    if (p→right == NULL)
        return 1 + findsomething(p→left);
    else
        return 1 + min(findsomething(p→left), findsomething(p→right));
}
```

What does the above function try to calculate?

A. Level of nearest leaf to root
B. Level of farthest leaf to root
C. Total number of nodes in left subtree of root
D. Total number of nodes in one of subtree of root whichever is having lesser nodes.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Suppose an int A is stored at virtual address 0xff987cf0, while another int B is stored at virtual address 0xff987d 98. If the size of a page is 0x1000 bytes, then A's physical address is numerically less than B's physical address. Assume both int A and int B belong program.

A. Always true
B. Always false
C. Sometimes true, sometimes false
D. Not enough information

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 1 | Penalty: 0 | Programming in C

If the 32-bit number given below is stored in little-endian at address 1000, what is the value of the byte at address 1002? Answer in decimal.

$$0001\ 0010\ 0011\ 0100\ 0101\ 0110\ 0111\ 1000$$

This number is the same as 12345678 in hexadecimal.

Your Answer: | Correct Answer: 52 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 1 | Penalty: 0 | Compiler Design

A derivation of a string w of length n in a context-free grammar

A. must involve exactly $2n-1$ applications of rules for all n.
B. must involve at least n rules, but can involve an arbitrarily large number.
C. must involve at least n rules and at most 2^n rules.
D. can involve any positive integer number of rules.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Consider the following Context-Free Grammar (CFG) over the language $\Sigma = \{a, b, c\}$:

$$S \rightarrow aE \mid abQ$$
$$E \rightarrow cQ|cb|\varepsilon$$
$$Q \rightarrow aE \mid \varepsilon$$

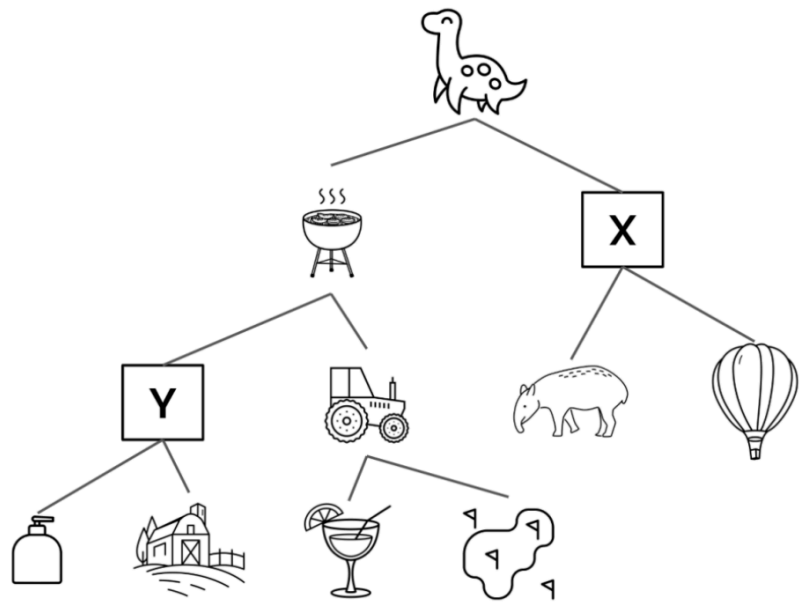
What are the Follow sets for each of the nonterminals?

A. Follow $(\mathrm{S}) = \mathrm{Follow}(\mathrm{E}) \neq \mathrm{Follow}(\mathrm{Q})$
B. Follow $(\mathrm{S}) = \mathrm{Follow}(\mathrm{E}) = \mathrm{Follow}(\mathrm{Q})$
C. Follow $(\mathrm{E}) = \mathrm{Follow}(\mathrm{Q}) \neq \mathrm{Follow}(\mathrm{S})$
D. Follow $(\mathrm{S}) \neq \mathrm{Follow}(\mathrm{E}) \neq \mathrm{Follow}(\mathrm{Q})$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Shown below is a binary Min Heap, where each symbol represents a node in the heap. Assume that each symbol represents a distinct value. Further, assume that $\mathrm{X} > \mathrm{Y}$.



Which of the following COULD be true about the relationship between the symbols?

A. $X >$

B. $X >$

C. $X <$

D. $X <$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

It is possible to create a tree that is both a binary search tree and a heap (max or min). What is the maximum number of nodes in such a tree?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Which of the following statements is TRUE about Huffman coding? In all options assume distinct probabilities for all items.

A. In Huffman coding, the item with the second-lowest probability is always at the leaf that is furthest from the root.
B. In Huffman coding, the item with the highest probability is always at a leaf that is the child of the root.
C. If the frequencies of the letters $a, b,$ and c are equal, then $a = 01, b = 10, c = 11$ is a valid Huffman code.
D. None of these

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ k & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. For which value of k does the system $A\mathbf{x} = b$ have a unique solution?

A. There is no such value for k.
B. $k = 0$
C. $k = -1$
D. $k = 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Which of the following statements is/are true?

I. $(a \rightarrow b)$ always equals $\overline{a} + b$
II. $(\overline{a} + b) + (a * \overline{b})$ is a tautology
III. $(a \rightarrow b) * (a * \overline{b})$ is satisfiable

A. III only
B. I and II only
C. II and III only
D. I, II, and III

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Numerical Type | Award: 1 | Penalty: 0 | Probability

Let X be a random variable (RV) that takes values -1 and 1, with $P(X = -1) = \frac{1}{2}$. What is the expected value $E\left[X^2\right]$?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Numerical Type | Award: 1 | Penalty: 0 | Combinatory

Let $\mathrm{A} = \{1, 2, 3, 4, 5, 6\}$. How many injective functions $\mathrm{f} : \mathrm{A} \rightarrow \mathrm{A}$ have the property that for some $x \in \mathrm{A}, \mathrm{f}(x) \neq x$?

Your Answer: | Correct Answer: 719 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X \in \{0, 1\}$ and $Y \in \{0, 1\}$ be two independent binary random variables. If $P(X = 0) = p$ and $P(Y = 0) = q$, then $P(X + Y \geq 1)$ is equal to

A. $pq + (1 - p)(1 - q)$
B. pq
C. $p(1 - q)$
D. $1 - pq$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Suppose $g(x)$ is a polynomial function such that $g(-1) = 4$ and $g(2) = 7$. Then there is a number c between -1 and 2 such that

A. $g(c) = 1$
B. $g'(c) = 1$
C. $g(c) = 0$
D. $g'(c) = 0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Let A be a regular language and B be some other language that we know nothing about. Suppose we find that $A \cup B$ is not regular. Which of the following can we say about B?

A. B is regular.
B. B is not regular.
C. B is CFL.
D. None of the above.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

In microprocessor systems with memory-mapped $\mathrm{I/O}$, which of the following is true?

A. Only $\mathrm{I/O}$ devices with internal memory can be interfaced.
B. $\mathrm{I/O}$ devices can be accessed using IN and OUT instructions.
C. Each $\mathrm{I/O}$ device can be addressed as a memory location.
D. Arithmetic and logic operations cannot be directly performed with the $\mathrm{I/O}$ data.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

The following table shows the edge weights in a complete graph on 6 vertices.

	1	2	3	4	5	6
1		8	13	5	3	6
2	8		9	14	13	2
3	13	9		17	19	13
4	5	14	17		12	3
5	3	13	19	12		7
6	6	2	13	3	7	

Calculate the cost of the Minimum Spanning Tree (MST) for the given graph using the table above.

Your Answer: | Correct Answer: 22 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider functions $f()$ and $g()$ which are positive increasing functions.

Which of the following(s) is/are correct?

A. If $f(n) = o(g(n))$ then $\log(f(n)) = o(\log(g(n))$, here o is small-oh
B. If $f(n) = \mathrm{O}(g(n))$ then $e^{f(n)} = \mathrm{O}\left(e^{g(n)}\right)$, here O is big-oh
C. If $f(n) = O(g(n))$ then $\log(f(n)) = O(\log(g(n))$, here O is big-oh
D. If $f(n) = o(g(n))$ then $e^{f(n)} = o\left(e^{g(n)}\right)$, here o is small-oh

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Suppose that on a certain system, every function call puts 10 bytes onto the stack plus another 4 bytes onto the stack for every integer formal parameter. Consider the following function definitions.

```
    Function walk( int n ) {
        if ( n <= 1 ) then return
        run( n, n / 2 )
    }
5.  Function run( int x, int y ) {
        if ( x >= y ) then walk( y )
    }
```

If n is a power of 2, then how much stack space must be available so that walk(n) can be called?

A. $14 * \log_2(n)$
B. $18 * \log_2(n)$
C. $32 * \log_2(n) - 18$
D. $32 * \log_2(n) + 14$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider two natural-valued functions $f : N \to N$ and $g : N \to N$. Which of the following statements cannot be true?

A. $f \in \Theta(g)$ and $g \in \Theta(f)$
B. $f \in \Omega(g)$ and $g \in \Omega(f)$
C. $f \in O(g)$ but $g \notin \Omega(f)$
D. $f \notin O(g), \Theta(g)$, or $\Omega(g)$, and $g \notin O(f), \Theta(f)$, or $\Omega(f)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the matrix $\begin{bmatrix} 5 & -1 \\ 4 & 1 \end{bmatrix}$.

Which one of the following statements is TRUE for the eigenvalues and eigenvectors of this matrix?

A. Eigenvalue 3 has an algebraic multiplicity of 2, and only one independent eigenvector exists.
B. Eigenvalue 3 has an algebraic multiplicity of 2, and two independent eigenvectors exist.
C. Eigenvalue 3 has an algebraic multiplicity of 2, and no independent eigenvector exists.
D. Eigenvalues are 3 and -3, and two independent eigenvectors exist.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A cache is having a 60% hit ratio for read operation and 80% hit ratio for write operation. Cache access time is 30 ns and main memory access time is 100 ns, 50% operations are read operations, and 50% are write operation. What will be the average access time for read operation (in ns)?

Your Answer: | Correct Answer: 70 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

In a computer with a cache memory interposed between the processor and the primary memory, the cache is fully associative cache. Let there be P locations(blocks) in primary memory and C locations(lines) in the cache.

On the average, how many different locations in primary memory map to a particular location in the cache?

A. 1
B. P
C. P/C
D. $2\,\mathrm{P}\,/\,\mathrm{C}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

Which of the following pairs of twos-complement two-byte integers, written in hexadecimal, would result in overflow if added? (Select all that apply)

A. 4650, 2340
B. FFED, FFFF
C. 787A, E3E0
D. 878A, E0E3

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Which of the following statements is/are true?

I. $A \cup (B - C)$ always equals $(A \cup B) - (A \cup C)$
II. $A \cap (B - C)$ always equals $(A \cap B) - (A \cap C)$

III. $A - (B \cap C)$ always equals $(A - B) \cup (A - C)$

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let G be a finite cyclic group with more than 2 elements. The number of distinct generators in G is?

A. Necessarily Even
B. Necessarily Odd
C. Zero
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Consider the collection of all undirected graphs with 10 nodes and 6 edges. Let M and m, respectively, be the maximum and minimum number of connected components in any graph in the collection. If a graph has no self-loops and there is at most one edge between any pair of nodes, then what is M^{m}?

Your Answer: | Correct Answer: 2401 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

A transaction schedule is serializable if its effect is equivalent to that of some serial schedule. Consider a bookkeeping operation consisting of two transactions $-T_1$ and T_2 - that are required to keep the sum $A + B + C$ unchanged. Which of the following pairs of transactions will always result in a serializable schedule?

I.

T_1	T_2
Lock A;	Lock B;
A = A - 10;	B = B - 20;
Unlock A;	Unlock B;
B = B + 10;	C = C + 20;

II.

T_1	T_2
A = A - 10;	Lock B;
Lock B;	B = B - 20;
B = B + 10;	Unlock B;
Unlock B;	C = C + 20;

III.

T_1	T_2
Lock A;	Lock A;
A = A - 10;	B = B - 20;
Unlock A;	Unlock A;
B = B + 10;	C = C + 20;

A. I only
B. II only
C. III only
D. I and II

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following table whose name is T:

A	B	C
1	10	100
2	10	10
3	40	100
4	30	200
5	25	90

The number of rows in the output of the following SQL query is?

```
select A from T as R
where not exists (
    select * from T
    where T.B > R.B and T.C > R.C)
```

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

If A is a regular language, then

A. A is closed under concatenation.
B. $\mathrm{A} \circ \mathrm{A}$ must be context-free.
C. A must be finite.
D. There exists an ambiguous context-free grammar for A.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Let L be a regular language. Which of the following is/are true?

A. If L is recognized by an n-state DFA, then $\Sigma^* \setminus \mathrm{L}$ is recognized by an n-state DFA.
B. If L is recognized by an n-state NFA, then $\Sigma^* \setminus L$ is recognized by an n-state NFA.
C. If L is recognized by an n-state DFA, then L^* is recognized by an ($n+$ 2) state DFA.
D. If L is recognized by an n-state DFA, then L^* is recognized by an ($n+$ 2) state NFA.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the following statements:

I. The set $\mathbb{Q} \times \mathbb{Z}$ is uncountable.
II. The set $\{f : f$ is a function from $\mathbb{N}$ to $\{0, 1\}\}$ is uncountable.
III. The set $\{\sqrt{p} : p$ is a prime number $\}$ is uncountable.
IV. For any infinite set, there exists a bijection from the set to one of its proper subsets.

($\mathbb{Q}$ is the set of all rational numbers, $\mathbb{Z}$ is the set of all integers and $\mathbb{N}$ is the set of all natural numbers)
Which of the above statements are TRUE?

A. I and IV only
B. II and IV only
C. II and III only
D. I, II and IV only

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

For any Boolean function, a Non-Essential Prime Implicant is a Prime implicant that has no element that cannot be covered by other prime implicants.

Which of the following expressions is obtained from a non-essential prime implicant for the given $\mathrm{K\text{-}map}$?

cd \ ab	00	01	11	10
00	1		1	1
01		1	1	
11			1	1
10	1		1	1

A. $bc'd$
B. $d'b'$
C. ad'
D. abc

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following languages is/are decidable?
(Note that in the cases, M refers to a Turing Machine, and $\langle \mathrm{M} \rangle$ is some encoding of M, and $\mathrm{L}(\mathrm{M})$ is the language recognized by M)

A. $\mathrm{L} = \{\langle \mathrm{M} \rangle | \mathrm{L}(\mathrm{M})$ is a context-free language $\}$.
B. $\mathrm{L} = \{\langle \mathrm{M} \rangle |$ when started on a blank tape, Turing machine M eventually halts in a final state $\}$.
C. $\mathrm{L} = \{\langle \mathrm{M}_1, \mathrm{M}_2 \rangle \mid \mathrm{L}(\mathrm{M}_1) = \mathrm{L}(\mathrm{M}_2)\}$.
D. $\mathrm{L} = \{\langle \mathrm{M}, w \rangle \mid \mathrm{M}\,\mathrm{accepts}\, w$ in less than 100 steps $\}$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Numerical Type | Award: 2 | Penalty: 0 | Programming in C

Assume the following C code is inside the main function and there are no syntax or runtime errors:

```
int A[] = {3, 1, 4, 9, 7, 2, 5, 8};
int *p = &(A[3]) - 2;
p += *p;
printf("%d", *p);
```

What will be the output printed by this program when executed?

```
int A[] ={3, 1, 4, 9, 7, 2, 5, 8};
int *p = &(A[3]) - 2;
p += *p;
printf("%d", *p);
```

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider the following C code. Where x and y are unknown integers and have been initialized to values greater than 1.

```
int i=1, j=0;
while (i<y) {
    j = j+i;
    i = i+1;
5.  if(i==x) break;
}
```

Which of the following statement(s) about the variables i and j must be true after execution of this program?

A. $(j = y(y-1)/2 \wedge (i = y)) \vee ((j = x(x-1)/2) \wedge (i = x))$
B. If $x > y$, then $i = y$
C. If $x > y$, then $i = y - 1$
D. $(j = y(y+1)/2 \wedge (i = y)) \vee ((j = x(x+1)/2) \wedge (i = x))$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Consider a three-link configuration $\mathrm{A\text{-}X\text{-}Y\text{-}B}$, where host A is connected to router X, router X is connected to router Y, and router Y is connected to host B. All nodes support IPv4.

- The link $A - X$ has MTU $= 1500$ bytes
- The link $X - Y$ has MTU $= 500$ bytes
- The link $Y - B$ has MTU $= 492$ bytes

Suppose host A is sending a 1000-byte packet to host B. Assume there is no packet loss due to bit errors or buffer overflow.

Also, assume that the IP header size $= 20$ bytes.

How many packets will flow over the $X - Y$ link? And how many will arrive at host B?

A. 3 packets over $\mathrm{X - Y}, 5$ packets arrive at host B.
B. 2 packets over $\mathrm{X - Y}, 4$ packets arrive at host B.
C. 5 packets over $\mathrm{X - Y}, 5$ packets arrive at host B.

D. 3 packets over $X - Y$, 4 packets arrive at host B.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Imagine a world where IP addresses are 4 bits long (not 32). Suppose a router uses longest-prefix matching and has the following forwarding table:

Prefix	Output Interface
10	A
1	B
x	C
default	D

Suppose that a packet with address 1001 is forwarded to output interface C. How many possible addresses will be forwarded to output interface B?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

A message of size $10,000$ bits (not bytes) is sent from a source node A to a destination node B passing through two routers $\mathrm{R1}$ and $\mathrm{R2}$. All three links on the path have a delay of $20\ \mathrm{ms}$. Node A has a transmission rate of $100\ \mathrm{bits/sec}$, $\mathrm{R1}$ and $\mathrm{R2}$ have a transmission rate of $1000\ \mathrm{bits/sec}$. Find the total sum (in seconds) of transmission and propagation delays in transmitting the file from A to B when message is broken into 10 packets each of size 1000 bits. Answer up to 2 decimal places.

Your Answer: | Correct Answer: 102.06 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Let $< \mathrm{P}_1, \mathrm{P}_2, \ldots, \mathrm{P}_n >$ be the ONLY safe state of a system. Then which of the following is/are ALWAYS true ?

A. If P_2 is the first process to request for some resources then P_2 always have to wait.
B. P_1's requests will always be satisfied without even checking if resulting state (after processing P_1's requests) is in safe state or not.
C. Processes requests will always be satisfied in the same order as safe state sequence and we can not satisfy any out of order process request.
D. We can never satisfy any request coming from P_n without satisfying request from P_1.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Suppose there is a machine with 32-bit addresses and a two-level page table (in memory) such that the first 10 bits of an address is an index into the first-level page table and the next 10 bits are an index into a second-level page table. Suppose also that each entry in the page tables is 32-bits. How much minimum space is occupied in memory by the page tables for a process that has $64\ \mathrm{MB}$ of actual virtual address space allocated?

A. 4100 KB
B. 4096 KB
C. 68 KB
D. 72 KB

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following three threads and four semaphores.

$s1, s2, s3, s4$ are counting semaphores and x is a shared variable initialized as 1.

T 1	T 2	T 3
while (x $\neq$ 360){	while (x $\neq$ 360){	while (x $\neq$ 360){
P(s1);	P(s2);	P(s3);
P(s4);	P(s4);	P(s4);
x = x*2;	x = x*3;	x = x*5;
V(s4);	V(s4);	V(s4);
}	}	}

The goal is to initialize the semaphores to guarantee the termination of the process, i.e., threads should not run infinitely.

Which of the following initial values will ensure the process terminates correctly?

A. $s1 = 3, s2 = 2, s3 = 1, s4 = 1$
B. $s1 = 2, s2 = 2, s3 = 2, s4 = 2$
C. $s1 = 1, s2 = 1, s3 = 1, s4 = 3$
D. $s1 = 3, s2 = 1, s3 = 1, s4 = 2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

Consider the following grammar:

1. $\mathrm{S} \rightarrow \mathrm{CC}$
2. $\mathrm{C} \rightarrow \mathrm{Cc}$
3. $C \rightarrow d$

It is know that given grammar is $\mathrm{LR}(1)$.

Which of the following statements is CORRECT about parsing the string "dcd" using this LR(1) parsing table?

A. The string "dcd" will be accepted after a total of three shift operations and two reduce operations.
B. The string " dcd " will be accepted after a total of three shift operations and three reduce operations.
C. The string "dcd" will be accepted after a total of three shift operations and four reduce operations.
D. The parser will reject the string "dcd" due to a conflict in the parsing table.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A small disk drive with 80 cylinders (numbered 0 through 79) receives this batch of cylinder requests:

$$4, 16, 3, 43, 60, 2, 79$$

The head is initially at cylinder 40 and was headed in the direction of higher cylinder numbers (away from the spindle) when the batch of requests arrived. Assuming that seek time is proportional to seek distance (measured as number of cylinders traversed), which of these algorithms will result in the highest total seek time?

A. SSTF
B. SCAN
C. LOOK
D. C-LOOK

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a single-issue processor with an in-order five-stage pipeline (IF, ID, EX, MEM, and WB) and with the following characteristics.

- The register file can not be written and then read in the same cycle.
- The pipeline does not have other bypassing/forwarding hardware.

The first stage corresponds to instruction fetch from memory, the second corresponds to data load from registers, the third corresponds to ALU operation, the fourth corresponds to memory access, and the fifth corresponds to register update. Each stage occupies one processor cycle.

Suppose that a certain instruction in a program needs to read from a register that is written by the previous instruction in the program (read-after-write data hazard). How many cycles, if any, are wasted due to a pipeline stall?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Let $\mathrm{S} = \{0, 1\}$ be an alphabet of two symbols. For any string w, let $l(w)$ denote the length of w. Which of the following languages over S is/are not regular?

A. $\mathrm{L} = \{uv \mid u, v \in \mathrm{S}^*, l(v) = 2.\, l(u)\}$
B. $\mathrm{L} = \{ww \mid w \in \mathrm{S}^*\}$
C. $\mathrm{L} = \left\{\left(0^k 1^k\right)^k \mid k \geq 1\right\}$
D. $\mathrm{L} = \{ww \mid w \in \{0\}^*\}$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

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